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Primate Models of Separation-Induced Depression

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Picture a healthy, happy young primate living in a familiar, nonthreatening environment, and the image is likely to include an animal that is bright-eyed, alert, and active. One expects to see energetic activity characterized by bold locomotive outbursts often engaged in with seemingly reckless abandon. Except when its social group is actively in motion or responding to some external threat, play and exploration seem to fill the young monkey's daylight hours, with only intermittent pauses on the mother's ventrum to nurse or simply seek out maternal contact comfort. Vigorous wrestling and chasing with its peers, intermixed with gymnastic exercise play, are typical of the youngster's day. No object, social or otherwise, fails to elicit its immediate and rapt attention. Its energies seem boundless.

This description aptly describes what one sees in many settings in the laboratory and field when observing the behavior of the normal young of any of the social primates. But now consider another description of a similar-aged young primate, also in sound physical health. This young animal sits, quite immobile, hunched over into a collapsed posture, its head below its knees. Its arms are tightly wrapped around itself, and its toes are curled tightly inward. The youngster's eyes are closed even though this is the active part of the daylight hours. One of its digits, or perhaps a bit of the fur of its arms, is held in the mouth and sucked. A loud noise produces only a slow lift of the head, which is dropped again almost immediately. A familiar peer, tugging on its arm or bouncing off its back and then looking back before taking off in a play initiation, fails to elicit even a partial response. Nothing in its environment arouses or holds its attention. When food is presented, the youngster may either eat to excess or fail to respond to the food at all. The pattern persists for many days.

The infant or juvenile described here is manifesting the pattern of behaviors that have been described as "depression" in the nonhuman primate infant following some type of maternal loss. Not all subjects show the com-

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plete pattern, nor each of its elements at the same level of intensity. Under some conditions, the depression may be seen only briefly during the day and for only a few days at a time. Under other conditions, the pattern, at full intensity, may last for several weeks and even be elicited several months after separation from the mother. It is this depressive pattern in nonhuman primate infants, and our understanding of the factors that appear to govern its emergence, which serves as the core of this article.

The symptoms of "depression" described for the young monkey separated from its mother are, along with accompanying physiologic changes, very similar to symptomatology observed in depressed human patients. Furthermore, separation and loss have been implicated in development of depression and related medical illness for many years[8] and have continued to be central concepts in the investigation of the etiology of depression.[12, 43] Based on the similarity of symptoms, the acknowledged contribution of separation experiences to the onset or exacerbation of depressive disorder in humans, and the reversal of depressive affect and behaviors in both separated monkeys and human patients with, for example, electroconvulsive therapy (ECT) and tricyclic antidepressants,[45] a number of investigators[17, 30, 47] maintain that social separation in nonhuman primates is an appropriate animal model for human depression.

The importance of the study of human depression related to separation and loss is becoming increasingly apparent as the evidence indicates a significant role for bereavement in the development not only of depression and related psychiatric disorders, but for increased morbidity and susceptibility to other medical illness in patients experiencing traumatic loss.[31]

ANIMAL MODELS IN THE STUDY OF HUMAN PSYCHOPATHOLOGY

The use of animal models in the study of human psychiatric disorders has not been as readily accepted as those involved in other areas of medical research. However, the past several years have seen a burgeoning interest in carefully refined animal models of psychopathology, as the significant contributions of these models to such areas as neurochemical substrates of schizophrenia and affective disorder[21] have become apparent. The value of animal models of human pathologies of any type is most obvious when taking into account the ethical difficulties connected with the experimental study of humans. It is not possible to conduct prospective studies, and the investigations of the etiologies of psychiatric disorders must of necessity be retrospective. The assessment of therapeutic intervention is also difficult, since treatment factors are frequently confounded, controls are often unavailable, and the proper evaluation of therapies based on follow-up studies is often impossible. Many of these problems can be overcome with the use of well-designed and controlled animal studies; comprehensive collection of data, prospective studies, follow-up studies that can include life-span experiments in many species, and the assessment of various treatments and interventions can all be accomplished using appropriate animal modeling techniques.

The establishment of a rationale for the use of animal models in studies

of psychopathology does not necessarily provide evidence for generalizability or validity, and this is obviously a very crucial issue. Suomi[45] reviews criteria for the validity of animal models and concludes that they are "largely focussed on known characteristics of the human disorder being modelled." The model is considered valid to the extent that "(a) the objectively defined symptoms of the human disorders are mirrored in the animals under study; the same should be true for any known physiological or biochemical mechanisms underlying the modelled disorder, (b) known predisposers and direct causes of the dysfunction can effectively and consistently produce a parallel dysfunction in the animal species, and (c) procedures or treatments that reverse the disorder in humans can have therapeutic effects for the animals in which the disorder has been reproduced."

In accordance with the increasing interest in valid animal models of psychopathology, numerous research efforts using a vaiety of pharmacologic, conditioning, and environmental manipulation techniques have contributed significantly to the development of models of psychiatric disorders.[44, 46, 48] These include attempts to produce affective disorders,[25, 34, 40] psychosis,[10] and various conditioning models of anxiety.[19] Early documentation of observations of behavioral disturbances considered analogous to human anxiety syndromes was provided by Pavlov,[28] who reported severe distress in his experimental animals during difficult discrimination trials; "experimental neurosis" in laboratory monkeys during studies in which the animals were subjected to the close juxtaposition of a frightening stimulus and a desirable food reward was described by Masserman and Pechtal.[24]

With the advent of anxiolytic and antidepressant drugs, animal models of anxiety and depression have, in recent years, brought behavioral and pharmacologic induction, and subsequent assessment and treatment, of these conditions into psychiatric research.[9, 15, 27] Nonhuman primate models of depression, involving maternal and other social separation and loss, are clear examples of the development of behavioral models emerging from a solid theoretical foundation in the symptomatology and etiology of a human psychiatric condition and the incorporation of new findings related to potential physiologic and neural substrates of depression into existing theory.[31, 34]

MATERNAL SEPARATION AND DEPRESSION IN NONHUMAN PRIMATES

Maternal separation as the basis of primate models of depression came into use in the 1960s,[14, 35, 38, 39] when the pioneering investigators in this field became interested in determining whether Bowlby's[2, 3] observations of a protest-despair-detachment sequence of reactions to separation in human children could be produced in young monkeys. Harlow's group obtained results that essentially supported the theoretical construct of a "separation syndrome . . . as biologically basic as the activation of basic component instinctual responses. . . ."[39] However, the monkey separation-based model of depression has most consistently produced a two- rather than three-stage

sequence of responses, in which protest and despair, but not detachment, have been repeatedly demonstrated in separated infants.

There have been a number of theoretical interpretations of the behavioral response to separation observed in nonhuman primates. Kaufman and Rosenblum[17, 35] obtained profoundly depressed behavior in pigtail macaque infants following maternal separation. They characterized this depressive reaction in their subjects in terms of Engel's theory of conservation withdrawal.[7] Other theoretical perspectives that have been considered include Bowlby's theory of primary separation anxiety, which was the theoretical foundation for some of the first separation work,[38] Spitz's "anaclitic depression,"[11, 41] Seligman's theory of learned helplessness,[40] and the opponent process theory of motivation,[42] which figures prominently in the work of Mineka and Suomi.[26]

Although the early studies of separation-induced depression used mother–infant separation designs, paradigms have been developed that involve other types of social separation (such as infant–surrogate and peer–peer). There are various reasons for the emergence of these other separation paradigms, ranging from the expense of using primate mother–infant pairs (compared with repeated separations of peers, for example) to the potential conceptual importance of examining similarities and differences in the reactions to separation in a variety of attachment types.

However, the rationale for the use of mothers and their infants in the induction of depression through social separation is a strong one. The maternal–infant bond is likely to be the strongest and perhaps the "purest" attachment bond the infant will experience, and it is probably prototypical of all other attachments formed later in life.[31] Problems with the maternal–infant separation model include the confounding of age with effects of separation on the infant when repeated separations are used. Developmental changes are rapid and profound in primate infants and may significantly affect the consequences of separation from mother. In addition to the potential effect of age at separation, the importance of the interruption of a number of regulatory (biologic rhythms), nutritional, and social buffer roles in which the mother serves may contribute to separation reactions in the infant.

In the course of normal primate development, the infant monkey is dependent on the mother for nutrition, physical support, and protection for a significant period of time; in the case of macaques, infants are initially held continuously by their mothers, so that access to nursing is constant and the infant is carried everywhere. The pace of development and maternal style in terms of encouraging juvenile independence vary across and within species. Generally speaking, macaque infants are weaned between the ages of 4 and 8 months. Infants' fear and avoidance reactions do not develop for several months, so that they are completely dependent on the mother for protection from external threat during the first months of life. Even after weaning is complete and independent social functioning is achieved by the juvenile monkey, the mother–offspring bond is close and continues through the subsequent birth and rearing of siblings, which occurs every 1 to 2 years under normal breeding conditions. Macaques do demonstrate negative emotional responses to the birth of siblings, and tran-

sient depressive reactions can occur at these times of necessary physical separation from the mother as the newborn assumes a position of primary contact.

The nature of mother–infant attachments has been characterized by Bowlby[4] in terms of security of attachment, determined by the extent to which the attachment figure is "available and accessible."[26] Attachments can be described, then, as "relatively 'anxious' or 'secure,' with those subjects at the anxious end of the continuum more likely to show severe reactions to loss."[37] Species comparisons along this dimension have proved useful in terms of the subsequent reactions of infants to separation from their mothers. Bonnet and pigtail macaques present a particularly interesting comparison of differences in the social context of rearing, which, in turn, appears to influence the nature of mother–infant attachments and reactions to separation.[33, 35]

Pigtails form less cohesive groups than bonnets, in which individuals tend to isolate themselves from other group members, and the tension level is often marked. Infants are guarded closely by their mothers, who prevent contact between their offspring and other group members and discourage attempts by the infants to leave close proximity during the early months of life. However, as the time of rebreeding approaches (when the infant is approximately 4 to 5 months old), the pigtail mother begins aggressive, punitive weaning and removal of the infant from her body. This active period of physical rejection by the mother continues until the infant is 8 or 9 months of age.

In contrast, bonnet macaques quickly form cohesive, gregarious social groups in which members are usually in close contact, particularly while resting or sleeping. Long bouts of grooming and other affiliative activities are common among animals of both sexes and all ages. When infants are born, they are frequently handled and examined by group members with little interference from the mother. As the infant grows older, the mother, although attentive and protective, is typically passive regarding her baby's departures from and return to contact with her. Some active weaning on the mother's part occurs, but it is much more casual and less punitive than that observed in pigtail mothers.

Based on Bowlby's conceptualizations, it might be predicted that pigtails, because of the inconsistency of maternal response and limitations on their range of early social and environmental experience, would form "anxious" attachments and would consequently show more severe responses to loss and separation than the bonnets raised in a social environment in which "secure" attachments would be more likely to form. This, in fact, appears to be the case. Many studies have demonstrated pigtail infants' severe depressive reactions following loss,[17, 30, 32] whereas bonnets rarely exhibit the most negative symptoms and often seem to suffer few ill effects of the loss period.[16, 36]

Further support for the effects of the security afforded by the attachment bond has been obtained from observations of bonnet macaque mothers and infants living in environments in which the mothers were required to devote a substantial amount of time each day to searching for their food in a specially constructed foraging device. Control mothers, living in identi-

cal pens, were provided with ample amounts of food requiring no search or acquisition time. During the course of early development, the infants living in the foraging group appeared to be more independent of their mothers than control infants. They were apart from their mothers more frequently and for longer periods of time than their control counterparts, and although they were weaned more quickly than controls, they were not consistently rejected by their mothers. However, the normal equanimity of the bonnets was disturbed in the foraging group, with more overt hostility and less affiliative behavior than no-work controls.[37]

When mother–infant pairs were separated, the infants of the foraging mothers showed the most severe depressive reactions and were the slowest to recover even nominally normal functioning. It could be suggested that the mothers required to spend a fair portion of their day preoccupied with finding food were frequently not as available as control mothers fed ad lib; observational records show that the foraging mothers were generally working with the infants out of contact. The necessity for the mother to divide her time between her infant and food acquisition, coupled with the increased tension among the adults in the foraging group, may have left these infants less securely attached to their mothers and less able to learn coping skills necessary to deal with the requirements of the environment.[29]

Another type of foraging environment in which the feeding task conditions change repeatedly over time produced an increasingly aggressive social group with mothers whose rejection of their infants seemed positively correlated with their own rising levels of distress. The infants, forced to spend considerable amounts of time apart from their mothers, showed increasing disturbance. In some cases, the complete depressive pattern was shown; this is ordinarily not as common an occurrence in bonnets, the subjects of this study, as in other macaque species studied.[36]

It may be apparent by now that the reaction to loss in nonhuman primates, as in humans, is not fixed and rigid, but rather subject to individual (constitutional and experiential), genetic (that is, species), and environmental influences, some of which have been alluded to thus far. Before going on with a discussion of the types of factors that can shape response to loss in animal subjects, specific features found within the range of responses will be briefly reviewed.

BEHAVIORAL AND PHYSIOLOGIC RESPONSE TO SEPARATION

One aspect of the behavioral response to separation has been described in the opening paragraphs of this article: that is the despair phase, in which the behaviors manifested by the young monkey appear to be a depressive reaction to loss. Despair is typically preceded by the protest phase, the initial response to separation, which is characterized by immediate frantic activity including locomotion and vocalizations. These activities appear to reflect attempts to communicate with and locate the missing attachment figure. The agitation will often subside into the depressive despair reaction described earlier, which, with prolonged separations, will subside as well. The occurrence and time course of these phases can de-

pend on a number of mitigating factors. The third phase—detachment—may or may not occur in monkey separation studies. In this phase, the infant or juvenile shows an emotional detachment from its mother upon reunion; this phase has been observed in human children when in the presence of caregivers from whom they have been separated.

In recent years, the physiologic correlates of behavioral responses to separation have been studied, usually with the focus on the output of the hypothalamic–pituitary–adrenal axis (plasma cortisol), neurochemical measures (urinary and cerebrospinal fluid [CSF] catecholamines and their metabolites), and physiological measures such as heart rate (HR) and electroencephalographic activity.

Adrenocortical activity and its relationship to behavioral symptoms has been the subject of numerous separation studies, with a partial goal of identifying reliable biologic markers for anxiety and depression through the use of primate models. It has been found repeatedly that maternal separation in infant macaques results in a marked rise in cortisol over baseline levels in 5 to 8 minutes, which continues going up for a number of hours as the separation is maintained.[45] In squirrel monkeys, a species in which cortisol output is normally extremely high relative to other primates, including humans, the rise in cortisol can continue for days after the separation begins.[23] If reunion with the mother occurs within approximately 20 minutes of separation, cortisol levels will normalize within a very short time (under 8 minutes usually), illustrating rather dramatically the mother's ability to abolish separation distress at both the behavioral and physiologic levels. If the separation is maintained, macaque infants' cortisol levels will begin to drop after 2 to 4 hours, although preseparation levels may not be attained for several days.[6] Cortisol levels do not always correlate precisely with behavioral changes observed in protest and despair phases of separation response.[23] Studies have shown that under some circumstances, behavioral and adrenocortical changes can occur apparently inversely, and one response can lag substantially behind the other.[23] These apparent discrepancies in behavioral and physiologic response to separation distress present an interesting area for future study in terms of when the systems operate separately and when they are more closely associated.

Levels of CSF (as opposed to urinary) catecholamine and indoleamine metabolites can provide a good indication of brain neurotransmitter activity during social separation. Kalin and associates,[16] for example, have shown that CSF levels of norepinephrine and the catecholamine metabolite homvanillic acid (HVA) are significantly reduced following separation in monkeys who show depressive reactions, but remain at preseparation levels in subjects who do not respond to separation with depression. Because CSF samples can be easily obtained in monkey subjects using a cisternal (rather than lumbar) tap, it may be possible to obtain significant information about neurotransmitter dysfunction hypotheses that figure prominently in many biologic theories of depression.[1]

Reite and colleagues[30] have studied psychophysiologic changes in pigtail infants separated from their mothers. They have found that immediate increases in heart rate (HR) and body temperature (BT) are typical of the initial reaction to separation and will persist into the first night of separa-

tion in all subjects. Individual differences emerge as the separation is maintained, however, with some subjects showing sharp decreases in HR and BT, so that after several days of separation, these values are lower than at preseparation baseline. Other subjects return to baseline levels, whereas still others show HR fluctuations, including marked HR arrhythmia, if the separation is not terminated.

Electroencephalographic recordings indicate major disruptions in sleep patterns at night in separated infants. During the first separation night, sleep latency increases dramatically, as do arousals, whereas total sleep time, slow wave sleep, and REM frequency and duration drop to below baseline levels. With prolonged separation, arousals and REM disturbance persist, but other aspects of the sleep pattern return to normal; individual differences in sleep disruption are significant and become more marked as the separation is prolonged.[30] It is interesting to note that the sleep disturbances reflect profound physiologic effects of separation at a time when behavioral reactivity is minimal (during sleep). A "rebound" effect can occur at reunion, so that there are frequently increases above baseline levels in total sleep time, REM, and BT; usually baseline levels are attained within a week following reunion, but HR arrhythmia may persist for weeks in some subjects.[31]

Reite's group has also documented disturbances in immune responsivity of separated infants, indicating deficiencies that can persist for weeks,[22] a phenomenon that has been observed in bereaved spouses.[5]

FACTORS INFLUENCING THE NATURE OF THE RESPONSE TO SEPARATION

Extensive research into the factors in the social and physical environment, both pre- and postseparation, have shown clearly that the form and intensity of individual response to loss can be influenced profoundly by several variables. Hinde and colleagues have investigated the effects of removing the mother from the group, leaving the separated infant with familiar conspecifics, versus removing the infant from the mother and group to be housed alone in a novel setting during the separation.[13] Infants removed to isolated housing during separation displayed agitation throughout separation but showed little depressive reaction; at reunion, behavior returned to normal preseparation activities within a short time. However, infants remaining in the group while their mothers were removed showed depressive responses to the separations and had much more difficulty adjusting to their mothers' return at reunion.

The effects of the demands of the physical environment on subsequent separation responses have been described in the discussion of the series of foraging demand manipulation studies conducted in this laboratory. Of particular interest here are the multiple levels at which an aspect of the physical environment affects overall group tension levels, as well as the nature of the maternal–infant relationship and presumably subsequent attachment, all of which may contribute significantly to the context in which the separation occurs and ultimately determine the severity of response. Such infant variables as the development of coping skills, environmental mastery, and

social functioning may, in fact, be determined in part by the nature of the environmental demand and influence the manner in which separation is dealt with by the infant.[29, 37]

Another type of influence on reaction to separation has been studied by Kraemer and McKinney,[21] that is, differential response to loss following various pharmacologic treatments during separation and reunion periods, using short-term, repetitive peer separations in juvenile macaques. Pretreatment response to these separations has generally been fairly mild. However, with the administration of very low doses of alpha-methyl-paratyrosine (AMPT), a catecholamine-depleting drug, during separations, severe depressive reactions occurred. However, there were no behavioral consequences of administration of AMPT during reunion. The fact that extremely low levels of AMPT produced dramatic depressive symptoms during separation indicates that AMPT and social separation act in a "highly synergistic fashion,"[45] so that depression is produced in subjects for whom separation alone or low-dose AMPT treatment is not sufficient to achieve the same results.

It will be important in future studies to understand the nature of interactive factors in producing severe symptomatology, and the use of primate subjects, in which prospective studies involve the manipulation of a variety of environmental, social, and individual physiologic factors, may prove highly useful in understanding these complex interactions. Another intriguing and essential area of study that has been developed in recent years is that of individual predispositions, possibly genetically determined, which may interact synergistically with environmental variables to result in a particular type of behavioral and physiologic response to loss. For example, Suomi has been able to determine that certain early behavioral and psychophysiologic responses to novel stimuli in very young monkeys can serve as good and very stable predictors throughout development of whether an animal will be at high risk for fearful and anxious reactions in the face of novelty, as well as how the subject will respond to social separation later in life.[47] Measures obtained as early as 3 to 4 weeks of life have thus far been shown to be stable predictors of individual response. The ability to identify high- and low-risk individuals makes possible the use of prospective studies in which individual predisposition can be studied in a variety of environmental and experiential settings to determine the relative effects of external and individual variables in the development of various disorders that may be homologous to human conditions. A related goal is the investigation of ways in which alterations in any of these factors, including treatment interventions, may be able to alleviate the negative psychological and physical sequelae of severe reactions to stressors such as separation and loss.

REFERENCES

1. Akiskal HS, McKinney WT: Overview of recent research in depression. Arch Gen Psychiatry 32:285–305, 1975
2. Bowlby J: Grief and mourning in infancy and early childhood. Psychoanal Stud Child 15:9–52, 1960
3. Bowlby J: Separation anxiety: A critical review of the literature. J Child Psychol Psychiatry 1:251–269, 1961

4. Bowlby J: Attachment and Loss. Vol. 2. Separation. Basic Books, New York, 1973
5. Brown GW, Harris T, Copeland JR: Depression and loss. Br J Psychiatry 130:1–18, 1977
6. Coe CL, Levine S: Normal responses to mother–infant separation in non-human primates. In Klein D, Rabkin J (eds): Anxiety: New Research and Changing Concepts. New York, Raven Press, 1981
7. Engel GL: Anxiety and depression-withdrawal. The primary effects of unpleasure. J Psychoanal 43:89–97, 1962
8. Freud S: Mourning and Melancholia (1917). In Jones E (ed): Sigmund Freud. Collected Papers. New York, Basic Books, 1959
9. Friedman S, Paully GS, Rosenblum LA: A non-human primate model of panic disorder. Psychiatry Res, in press, 1987
10. Haber S, Barchas PR, Barchas JD: Effects of amphetamine on social behaviors of rhesus macaques: An animal model of paranoia. In Hanin I, Usdin E (eds): Animal Models in Psychiatry and Neurology. New York, Pergamon Press, 1977
11. Harlow HF, Gluck JP, Suomi SJ: Generalization of behavioral data between nonhuman and human animals. Am Psychol 27:709–716, 1972
12. Heincke C: Parental deprivation in early childhood. In Scott J, Senay E (eds): Separation and Depression: Clinical and Research Aspects. Washington, DC, AAAS, 1973
13. Hinde RA, Davies L: Removing infant from mother for 13 days compared with removing mother from infant. J Child Psychol Psychiatry 13:227–237, 1972
14. Hinde RA, Spencer-Booth Y, Bruce M: Effects of 6-day maternal separation on rhesus monkey infants. Nature 210:1021, 1966
15. Insel TR, Champoux M, Scanlan JM et al: Rearing conditions and response to anxiogenic drugs. Presented at the annual meeting of the American Psychiatric Association, Washington DC, May 1986
16. Kalin NH, Cohen RN, Kraemer GW, et al: The dexamethasone suppression test as a measure of hypothalamic-pituitary feedback sensitivity and its relationship to behavioral arousal. Neuroendocrinology 32:92–95, 1981
17. Kaufman IC, Rosenblum LA: The reaction to separation in infant monkeys: Anaclitic depression and conservation-withdrawal. Psychosom Med 29:648–675, 1967
18. Kaufman IC, Stynes AJ: Depression can be induced in a bonnet macaque infant. Psychosom Med 40:71–75, 1979
19. Kimmel HD, Brennan AF: Conditioning models of anxiety. Int J Psychol 16:371, 1981
20. Kraemer GW, Ebert MH, McKinney WT: Separation models and depression. In Angst J (ed): The Origins of Depression: Current Concepts and Approaches. New York, Springer-Verlag, 1983
21. Kraemer GW, McKinney WT: Interactions of pharmacological agents which alter biogenic amine metabolism and depression. J Affect Disord 1:35–54, 1979
22. Laudenslager ML, Reite M, Harbeck R: Suppressed immune response in infant monkeys associated with maternal separation. Behav Neurol Biol 36:40–48, 1983
23. Levine S: Comparative and psychobiological perspectives on development. In Collins WA (ed): Minnesota Symposium on Child Psychology. Vol. 15. Hillsdale, New Jersey, Lawrence Erlbaum Assoc, 1982
24. Masserman JH, Pechtal CT: Neuroses in monkeys: A preliminary report of experimental observations. Ann NY Acad Sci 56:253, 1953
25. McKinney WT: Animal models in psychiatry. Perspect Biol Med 17:529, 1974
26. Mineka S, Suomi SJ: Social separation in monkeys. Psychol Bull 85:1376–1400, 1978
27. Ninan PT, Insel TM, Cohen RM, et al: Benzodiazepine receptor-mediated experimental "anxiety" in primates. Science 218:1332–1334, 1982
28. Pavlov IP: Lectures on Conditioned Reflexes. New York, International Publishers, 1928
29. Plimpton E, Rosenblum LA: Ecological and temporal dimensions of maternal loss in nonhuman primates. In Bloom-Feshbach J, Bloom-Feshbach S (eds): The Psychology of Separation Through the Life-Span. San Francisco, Jossey-Bass, 1985
30. Reite M, Short R, Seiler C, et al: Attachment, loss, and depression. J Child Psychol Psychiatry 22:141–169, 1981
31. Reite M, Short R: Maternal separation studies: Rationale and methodological considerations. In Ethopharmacology: Primate Models of Neuropsychiatric Disorders. New York, Alan R. Liss, 1983
32. Rosenblum LA: Infant attachment in monkeys. In Schaeffer R (ed): The Origins of Human Social Relations. New York, Academic Press, 1971

33. Rosenblum LA: The influence of the social and physical environment on mother-infant relations. Ann Instituto Superiore Sanita 18:215–222, 1982
34. Rosenblum LA: Monkeys' responses to separation and loss. In Osterweis M, Solomon, F, Green M (eds): Bereavement: Reactions, Consequences, and Care. Washington, DC, National Academy Press, 1984
35. Rosenblum LA, Kaufman IC: Variations in infant development and response to maternal loss in monkeys. Am J Orthopsychiatry 38:418–426, 1968
36. Rosenblum LA, Paully GS: Variability of environmental characteristics and mother-infant relations. Child Dev 55:305–314, 1984
37. Rosenblum LA, Sunderland G: Feeding ecology and mother-infant relations. In Gandelman R (ed): Parenting: Its Causes and Consequences. New Brunswick, New Jersey, Rutgers University Press, 1982
38. Seay BM, Hansen EW, Harlow HF: Mother-infant separation in monkeys. J Child Psychol Psychiatry 3:123–132, 1962
39. Seay BM, Harlow HF: Maternal separation in the rhesus monkey. J Nerv Ment Dis 140:434–441, 1965
40. Seligman MEP: Helplessness. San Francisco, W.H. Freeman Co, 1975
41. Spitz RA: Anaclitic depression. Psychoanal Stud Child 2:313–342, 1946
42. Solomon RL, Corbit JD: An opponent-process theory of motivation: I. Temporal dynamics of affect. Psychol Rev 81:119–145, 1974
43. Stenbach A: Object loss and depression. Arch Gen Psychiatry 12:144–151, 1965
44. Suomi SJ: Animal models of human psychopathology: Relevance for clinical psychology. In Kendall P, Butcher J (eds): Handbook of Research Methods in Clinical Psychology. New York, John Wiley and Sons, 1982
45. Suomi SJ: Social development in rhesus monkeys: Consideration of individual differences. In Oliverio A, Zappella M (eds): The Behaviour of Human Infants. New York, Plenum Press, 1983
46. Suomi SJ, Seaman SF, Lewis JK, et al: Effects of imipramine treatment of separation-induced social disorders in rhesus monkeys. Arch Gen Psychiatry 35:321, 1978
47. Suomi SJ, Kraemer GW, Baysinger CM, et al: Inherited and experiential factors associated with individual differences in anxious behavior displayed by rhesus monkeys. In Klein D, Rabkin J (eds): Anxiety: New Research and Changing Concepts. New York, Raven Press, 1981
48. Treit D: Animal models for the study of anti-anxiety agents. A review. Neurosci Biobehav Rev 9:207, 1985

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